Velum sensor mat on hospital bed

VELUM

Prevention Before the Wound Forms

The world's first continuous, passive sensor mat that detects pressure injuries days before they become visible — equitably, affordably, and without clinical intervention.

$11B	2–5 Days	<$50	24/7
US ANNUAL COST	EARLIER DETECTION	TARGET UNIT COST	PASSIVE MONITORING

01

A System Built to Fail Patients

$11B+

The annual cost of treating pressure injuries in the US alone. Sub-epidermal tissue begins dying **2–5 days before any wound is visible** — yet standard hospital protocol relies entirely on periodic visual inspections. By the time a nurse spots a wound, irreversible damage has already occurred.



Reactive by Design

Visual inspections detect wounds only after they form. No passive, continuous monitoring system is deployed at scale, leaving caregivers with no early warning capability.



A Documented Equity Gap

Early-stage erythema is clinically invisible on darker skin tones. Current methods systematically fail patients of colour — a severe, documented racial disparity in patient safety.



Access Locked Behind Cost

Commercial SEM handheld scanners cost $3,000–$10,000 and require trained staff. Nursing homes and home care settings — where bedsores are most prevalent — are effectively excluded.



Scale of the Crisis

Pressure injuries affect over 12% of hospitalised patients globally. Hospitals face financial penalties, legal liability, and reputational damage when they occur — yet prevention tools remain inadequate.

02

Meet Velum



Zero-Setup Deployment

Slide the mat under the bedsheet. Power on. Done. No calibration ritual, no clinical training required, no patient interaction.

Continuous Impedance Mapping

A multi-zone electrode grid produces a full spatial impedance map every 5 minutes, tracking tissue health across all critical anatomical pressure zones simultaneously.

Days-Early Detection

Detects localised tissue oedema — the physiological precursor to injury — before irreversible surface damage occurs, giving caregivers a multi-day intervention window.

Instant Wireless Alerts

When impedance drops ≥15% in any zone, the onboard ESP32 pushes a BLE/Wi-Fi alert to the caregiver dashboard, identifying the exact anatomical zone at risk.

Equitable Across All Skin Tones

Electrical measurement — not optical — means Velum performs identically regardless of melanin concentration,

eliminating the racial disparity that plagues visual inspection.

03

The Technology



SENSING MODALITY

Sub-Epidermal Moisture (SEM) + Bioelectrical Impedance Spectroscopy (BIS)

MEASUREMENT CYCLE

Full spatial map every 5 minutes

1–100 kHz AC sweep · <1 mA safe micro-current

01

Kapton Polyimide Substrate

Biocompatible, flexible film with screen-printed silver & carbon conductive ink electrode grid — enabling low-cost mass manufacturing.

02

AD5933 Impedance Converter

Paired with Howland current source for stable AC injection. 16-channel analog multiplexer cycles through all anatomical zones.

03

ESP32 Microcontroller

Dual-core processor with built-in BLE 5.0 and Wi-Fi. Continuously compares zone readings to patient baseline calibration.

04

Medical-Grade Silicone + TPU Encapsulation

Fully washable. Survives hospital-grade disinfection. Breathable membrane maintains skin microclimate.

05

Future: On-Device ML

Time-series impedance data will train predictive models to forecast injury risk before the threshold is even breached — building a proprietary clinical data moat.

04

Market Opportunity

TAM — TOTAL ADDRESSABLE MARKET

$11B+

Annual US cost of treating pressure injuries. Global wound care market exceeds $20B.

SAM — SERVICEABLE ADDRESSABLE MARKET

$3.2B

ICUs, long-term care facilities & home care in North America and Western Europe — highest-risk patient populations.

SOM — SERVICEABLE OBTAINABLE MARKET

$180M

12%+

Hospitalised Patients Affected

Pressure injuries affect over 12% of all hospitalised patients globally, with rates rising in ageing populations. The problem grows every year.

3 Settings

Addressable Care Environments

Hospital ICUs, nursing homes, and home care settings all share the same unmet need — and none have an affordable continuous monitoring solution today.

~$43,000

Cost to Treat a Single Stage IV Wound

First 5 years: 500 facilities × avg. 360 beds × $1/day

Hospitals pay an average of $43,000 per pressure injury episode. One prevented injury per quarter pays for Velum for an entire ward for years.

05

Competitive Advantage

CRITERION	VISUAL INSPECTION	HANDHELD SEM SCANNERS	PRESSURE-RELIEF MATTRESSES	✦ VELUM
Detection Timing	After wound visible	Periodic only	No detection	**2–5 days early**
Continuous 24/7	✗	✗	✗	✓
Equitable (All Skin Tones)	✗	~ Partial	✗	✓
Unit Cost	Staff time only	$3,000 – $10,000	$2,000 – $8,000	**< $50**
Patient Disruption	Moderate	High	None	**None**
Viable Beyond ICU	~ Limited	✗	~ Partial	✓
Clinical Data Generation	✗	✗	✗	✓ **Continuous**

The data moat: Every Velum deployment generates continuous, longitudinal bioimpedance time-series data tied to clinical outcomes — a dataset that does not exist anywhere today. This underpins the machine learning layer and creates a compounding competitive advantage that deepens with each new patient monitored.

06

Business Model & Roadmap



Hardware + Subscription

Low-cost mat hardware sold or leased to facilities. Monthly SaaS subscription per bed for dashboard, analytics, and alert management.

~$1/bed/day



Institutional Sales

Hospital networks and nursing home chains purchase fleet licenses. Procurement driven by compelling ROI: one avoided injury funds an entire ward.

B2B Enterprise



Data & Predictive Analytics

Future premium tier: ML risk-scoring engine trained on proprietary dataset. EMR integration for automated documentation and population analytics.

Premium Tier

PHASE 1 · 2026

Prototype & Validation

Bench prototype, IRB ethics approval, validation against commercial SEM scanners in a single ICU pilot.

PHASE 2 · 2027

Clinical Pilot

Multi-site pilot with 3–5 hospital partners. Regulatory pre-submission (FDA Class II pathway). First clinical dataset established.

PHASE 3 · 2028

Market Entry

FDA 510(k) clearance. First commercial deployments in US ICUs and nursing homes. Seat-pad variant for wheelchair users launched.

PHASE 4 · 2029+

Scale & Intelligence

ML risk-scoring layer live. European CE mark. Expansion into home care & domestic settings. EMR integrations with Epic & Cerner.

07

Traction & Milestones

Validated

Core Sensing Mechanism

Bench testing with AD5933 + conductive electrodes on agar gel tissue phantom demonstrates clear impedance shifts correlating with compression — confirming the fundamental detection principle.

5 Min

Full-Body Impedance Cycle

16-channel multiplexer architecture confirmed capable of completing a complete spatial scan across all anatomical zones within the 5-minute target interval.

<$50

Confirmed BOM Target

Component-level bill of materials analysis using screen-printed Kapton + AD5933 + ESP32 architecture confirms sub-$50 unit manufacturing cost at scale is achievable.

Award Pipeline

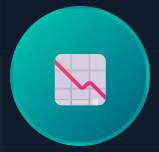
Recognition & Competitions

- James Dyson Award 2026 — Healthcare category entry
- Create The Future Design Contest — SDG3/10 entry
- Entrepreneurship World Cup — Active submission

- Falling Walls Foundation — Innovation pipeline

08

Why Now



Printed Electronics

Cost Inflection Point

Conductive ink screen-printing on flexible substrates has crossed a commercial viability threshold — enabling the sub-$50 unit cost that makes Velum's business model possible for the first time.



Low-Power IoT

ESP32 + BLE 5.0 Maturity

The ESP32's combination of dual-core processing, BLE 5.0, Wi-Fi, and sub-$5 cost makes continuous wireless monitoring viable in a disposable or semi-disposable form factor.



Healthcare Equity Push

Regulatory & Social Tailwind

CMS value-based care penalties, growing regulatory pressure on health equity, and documented racial disparities in pressure injury outcomes create a policy tailwind for equitable monitoring technology.



Ageing Global Population

Structural Market Growth



Hospital Financial Pressure

CMS Non-Reimbursement



Clinical AI Readiness

Data Infrastructure Matured

The WHO projects 2.1 billion people over age 60 by 2050. Immobility-related conditions including pressure injuries will grow proportionally, expanding the addressable market every year.

Since 2008, CMS does not reimburse hospitals for hospital-acquired pressure injuries. Each avoided wound is a direct financial benefit — making Velum's ROI case immediate and measurable.

Hospital systems have invested heavily in EMR infrastructure and clinical AI acceptance has grown significantly — making a data-generating device like Velum easier to integrate and sell than ever before.

09

The Ask

$750K

Pre-Seed Round — 18-month runway to clinical validation and regulatory pre-submission.

40%	Clinical Pilot & IRB Study
25%	Hardware Engineering & Prototyping
20%	Regulatory Pathway (FDA Pre-Sub)
15%	Team & Operations

Velum's mission is to make continuous pressure injury prevention a universal standard of care — in every ICU, every nursing home, and every home where someone lies still for too long.

Every day without Velum, sub-epidermal tissue is breaking down undetected in thousands of wards around the world. We intend to change that — starting now.

Get in Touch

velum@founders.vc · velum.health

$11B+	<$50	2–5 Days
MARKET SIZE	UNIT COST	EARLIER DETECTION